

PROTOKOL

Quality Traceability for Public Infrastructure
Technical Product Document — v2.4

Document	Technical Product Document (what to build, why, and what we found)
Version	v2.4 — supersedes v2.3. Post-audit fixes: corrected the API contract examples in Section 8 to match the real slump range and cylinder count (they still showed placeholder values), marked questions 3 and 7 in Section 13 as answered by Snehil, and added three more confirmed competitors (Calidad Cloud, ObraLink, Bildin) to Section 5.3.
Date	September 2026
Written by	Kenny Garamendi — Product
For	Snehil Sahay — Engineering
Market	Public road construction, Peru
Status	Pre-build. Nothing has been coded yet.

Read Section 2 first. Without it, the rest of this document is just words.

1. EXECUTIVE SUMMARY

We have privileged access to a live public infrastructure project in Peru — a S/ 41.9M (≈ USD 11M) road construction contract — including its full document archive, its internal coordination channel, and its quality manager. We spent six weeks analyzing 487 files and the complete team chat history.

What we found is a documentation system that is legally mandatory, financially blocking, and completely broken. Quality records that the law requires are being kept in paper notebooks, WhatsApp messages, and Excel files that contradict each other. Without those records, the contractor cannot get paid, and the quality engineer personally carries legal liability for seven years after delivery.

Protokol is the system that fixes this. It captures quality data at the point where work happens, validates it against the legally binding technical standard, and produces the document package the state requires — automatically.

Why this is worth building

Factor	Evidence
The problem is real and documented	We have the quality manager writing in all caps in the team chat: "AS OF TODAY WE ARE NOT MANAGING TO COMPLETE THE PROTOCOLS"
The market is large	2,741 public works projects are currently stalled in Peru, representing over S/ 67 billion in committed investment. Documentation failure is a leading cause.
We have insider access	Full access to a live project's document archive, team chat, and processes — most competitors build blind
There is a paying trigger	Missing quality documentation blocks the monthly payment certificate. This is not a nice-to-have.
We have a commercial channel	A civil engineer with sector reputation and institutional contacts is ready to sell this using his name and network

Who is involved

Person	Role	Relationship
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Kenny Garamendi	Product, research, go-to-market	Founder
Snehil Sahay	Engineering, architecture, implementation	Founder — full technical authority
David Valdez Ochoa	Domain expert and commercial channel. Civil engineer, currently Quality Specialist on the pilot project. Has sector contacts and institutional reach.	Not a founder. Commercial partner — terms to be defined separately.

Funding for Phase 0 is expected to be disbursed in milestones rather than as a lump sum — an initial payment to start, a second payment partway through development, and the remainder on delivery of the usable version. Exact percentages and dates are still being finalized with David's side; this document will be updated once they are fixed.

► *Snehil: plan your own time and invoicing around milestone payments, not a single up-front transfer. I will confirm exact amounts and dates before Phase 0 starts.*

2. HOW PERUVIAN PUBLIC CONSTRUCTION ACTUALLY WORKS

► *Snehil: this section exists because you cannot design this system correctly without understanding the physical and legal process it models. Read it slowly. Everything else depends on it.*

2.1 The physical process — how a road gets built

A concrete road is not poured in one go. It is built in horizontal layers, and each layer must be approved before the next one can go on top. This sequencing is the core of the system we are building.

Step	What happens physically	What must be verified
1. Subgrade	The natural soil is cut, leveled and compacted to form the foundation	Compaction density, moisture content, elevation
2. Sub-base	A granular material layer (20cm) is spread and compacted	Compaction $\geq 100\%$ of lab reference, thickness, material certificate
3. Base	A second granular layer (25cm), higher quality	Same as sub-base, plus load-bearing test
4. Formwork	Wooden or metal molds are set to contain the concrete	Dimensions, alignment, elevation
5. Steel	Reinforcement bars are placed inside the formwork	Bar diameter, spacing, concrete cover
6. Concrete pour	Ready-mix concrete arrives by truck and is poured into the formwork	Consistency test on arrival, sample cylinders taken, truck ID logged
7. Curing	The fresh concrete is kept wet for days so it develops strength	Curing method and duration
8. Strength test	The sample cylinders are crushed in a lab at 7 and 28 days	Compressive strength must meet design value

Key insight for architecture: this is a state machine over physical segments, not a CRUD app over forms.

2.2 The players — and who actually wants this product

Actor	What they do	Incentive regarding Protokol
Field technician	Executes and measures on site. Sweat, dust, phone in pocket.	Neutral to negative. More work unless it is faster than what they do now.
Quality Specialist	Signs off that everything complies. Personally liable for 7 years.	STRONGLY POSITIVE. This is our user and our champion. David holds this role.
Site Resident Engineer	Runs execution. Needs the monthly payment certificate approved.	Positive. Missing documents delay his payment.
Supervisor / Inspector	Independent auditor appointed by the state entity. Verifies and objects.	Positive. Less time chasing paperwork.
State entity (Regional Government)	Owns the project and the budget. Audited by the Comptroller.	Positive at institutional level, slow at decision level.
Comptroller General	National audit body. Can sanction	Not a customer, but the reason the customer

	individuals.	needs us.
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On this pilot, the Quality Specialist confirmed the exact team structure in person. The Execution Team is Site Resident, Soils Specialist, Quality Specialist, Assistant, and Safety Specialist. The Supervision Team is Supervisor, Structures Specialist (only when the work includes a bridge or reinforced concrete), Quality Specialist, and Assistant. The Quality Specialist role sits in both teams, which is why it is the entry point for Protokol.

► *This structure is standard across the Peruvian construction industry, not specific to this project — confirmed against multiple job postings for these same roles. Design the technician/approver model around these five-plus-four roles, not a generic two-role hierarchy.*

The real people signing these roles today, confirmed from six signed protocols in the field: Ing. Edison Cuadros Garcia (Site Resident, CIP 302775), David Valdez Ochoa (Quality Specialist, execution side), Ing. Teodoro Manuel Huamancusi Quispe (Supervisor, CIP 53548), Ing. Roly Conocachi Huamani (Structures Specialist, supervision side, CIP 76843), Ing. Cristian Manuel Torres Salinas (Quality Specialist, supervision side, CIP 260873). At least one subcontractor (sub-base supply) has its own separate Quality Specialist signing its own protocols — the technician model should allow a protocol to be tied to a subcontractor scope, not only to the main execution team.

2.3 The legal and financial machinery

Three facts drive the entire business case. Internalize these:

Fact 1 — Quality documents gate the money

Contractors get paid monthly through a document called a valorización (payment certificate). It measures physical progress and requests payment. If the quality documentation backing that progress is incomplete, the certificate is objected and payment is delayed. We have the internal memos from this project proving this — the Site Resident formally requesting quality certificates "for evaluation, conformity and payment processing."

Fact 2 — Personal liability lasts seven years

Under Peruvian Civil Code Article 1784, if a structure shows serious defects or partial collapse, the builder is liable. For public works, that liability extends a minimum of seven years from formal acceptance. The quality documentation is the only defense. A quality engineer with no records is personally exposed.

Fact 3 — There is already a mandatory state system, and it is a wedge, not a competitor

Since June 2024, Comptroller Directive N° 017-2023-CG/GMPL governs projects executed directly by state entities. It requires the resident and supervisor to register execution information in INFOBRAS — the Comptroller's public works information system — replacing the old paper site logbook. Registration is mandatory under sanctionable responsibility.

► *This is our strongest commercial angle and it must shape the architecture: Protokol should feed INFOBRAS, not compete with it. Design the data model so that exporting to INFOBRAS format is straightforward.*

2.4 Glossary — terms used throughout this document

Spanish term	English meaning
Progresiva	Chainage. Distance along the road centerline, written as km+meters. '0+144' = 144 meters from start.
Paño	Slab panel. A rectangular section of pavement poured in one work session. The daily unit of work.
Vaciado	Concrete pour.
Encofrado	Formwork. The temporary mold holding fresh concrete.
Slump	Concrete consistency test. A cone is filled, lifted, and the mix subsidence is measured in cm. Checks workability.
f'c	Design compressive strength of concrete, in kg/cm ² . Peru uses kg/cm ² , not MPa or psi.
Probeta / Testigo	Concrete test cylinder, cast during the pour, crushed at 7 and 28 days.
Mixer	Ready-mix concrete truck.
Curado	Curing. Keeping concrete moist so it develops strength.
Subrasante	Subgrade. Prepared natural soil layer.
Sub base / Base granular	Granular sub-base and base layers.

Proctor Modificado	Modified Proctor. Lab test establishing maximum achievable soil density. Field compaction is expressed as % of this.
Cota	Elevation level, verified by survey equipment.
Protocolo	Quality protocol. The formal record proving an activity met standard. Our core artifact.
PPI	Programa de Puntos de Inspeccion (Inspection Point Program). The formal, internationally recognized name for a protocolo. Defines who inspects, which standard applies, and whether it is a hold point (needs external sign-off) or a witness point (internal check only).
Liberación	Release / sign-off allowing work to proceed to the next stage.
No Conformidad (NC)	Non-conformance. Formal record that something failed a criterion. Requires documented corrective action.
Dossier de Calidad	Quality dossier. The complete compiled documentation package. Required to close the contract.
Valorización	Monthly payment certificate based on measured progress.
Expediente Técnico	Master project document: drawings, specifications, quantities, budget.
Metrado	Quantity take-off.
Residente de Obra	Site Resident Engineer — runs execution.
Supervisor / Inspector	Independent supervising engineer appointed by the state entity.
Administración Directa	Direct execution. The state entity builds with its own staff rather than contracting a construction company.
EG-2013	Ministry of Transport road construction specification manual. Legally binding technical standard.
INFOBRAS	Comptroller General's mandatory public works information system.

3. WHAT WE INVESTIGATED AND HOW

This document is not based on assumptions about the market. It is based on primary evidence from a live project. Here is exactly what we analyzed:

Source	Volume	What it gave us
Project quality Drive (full archive)	487 files across 49 folders, 6.79 MB of extracted text	The official document taxonomy, all procedures, the master protocol tracker, technical specifications, budget
Team WhatsApp group	Complete history, 31 Jul – 27 Aug 2026	How work is actually coordinated, real frustrations, response patterns
Official memoranda	6 formal memos between Site Resident and Quality Specialist	Proof that quality documents gate payment; contract number and legal framing
Monthly quality report	Report N°01-2026, written by the Quality Specialist	Complete project history: 10 time extensions, budget growth, and the state of quality management on arrival
Master protocol tracker	84 rows of live quality records	Quantified data inconsistencies
Technical file	Reformulated budget, drawings, quantity take-offs	Real activities, materials and specifications
Concrete mix designs	Lab reports from the accredited testing lab	The actual validation thresholds for this project
In-person meeting with the Quality Specialist (12 Sep 2026)	One session, multiple attendees	Confirmed the exact team structure, the protocol release chain per element, offline signing behavior, and an explicit request to pilot all four protocols (survey, formwork check, steel, concrete) rather than three
Six real, signed field protocols (16 Sep 2026)	Topography, Formwork x2, Steel x2, Concrete — all four Phase 0 activities covered	Confirmed real f'c values (280 kg/cm ² pavement, 210 kg/cm ² walls/culverts), the actual signature roster with full names and CIP numbers, and that the two formwork/steel template codes we thought were competing versions are actually for different element types (structural vs. pavement), not a conflict to resolve

The pilot project

Attribute	Value
Project	Road transitivity improvement and expansion, section AY-728 to AY-729
Location	Ayacucho region, Peru
Length	2.38 km
Contract	N° 81-2026-GRA-SEDECENTRAL-OAPF
Owner	Regional Government of Ayacucho

Execution mode	Direct execution (Administración Directa)
Original budget	S/ 16.3M — reformulated to S/ 41.9M (+156%)
Original duration	240 days — extended to 1,325 calendar days
Time extensions	10 (AP01 through AP10)
Progress at analysis	53.19% physical, 68.37% financial
Scope	Rigid pavement, asphalt, a bridge, curbs, sidewalks, culverts, gravity walls, retaining walls

4. FINDINGS — WHAT IS ACTUALLY BROKEN

Every finding below is backed by a specific artifact we hold. These are not hypotheses.

Finding 1 — Quality records are not being completed, and everyone knows it

On 8 August 2026, the Quality Specialist wrote to the entire team group, in capital letters:

"AS OF TODAY WE ARE NOT MANAGING TO COMPLETE THE PROTOCOLS. I NEED YOU YOURSELVES TO BE PREPARING THE PROTOCOLS AND FILLING THEM IN."

No one replied. The next messages in the chat are unrelated photos. Earlier the same day another team member wrote that there were no protocol forms left and they needed to be printed.

► *Interpretation that matters for design: the bottleneck is not awareness. Everyone knows the protocols are needed. The bottleneck is friction at the moment of recording.*

Finding 2 — There are five parallel sources of truth for the same data

The same construction reality is recorded in at least five disconnected places, none of which reconcile automatically:

- The pour schedule (Excel) — which panel gets poured on which day
- The ready-mix delivery log (Excel, 137 rows) — one row per concrete truck
- The lab test template (Excel) — sample cylinders and results
- A physical paper notebook — cement bags consumed per structural element
- The master protocol tracker (Excel) — quality release status

Concrete evidence of divergence: we hold two files with the same name, Cronograma_Progresivas.xlsx, shared days apart in the same chat. For the same road section on the same day, one states 96.5 m³ of concrete required and the other states 39.9 m³. One of them is wrong, and nobody knows which.

Additionally, the same technical drawing exists in four formats: the native CAD file, a clean PDF exported from it, a photocopier scan of a printed copy, and a re-shared duplicate under a different filename. We verified two of these files are byte-for-byte identical with different names.

Finding 3 — Paper loses data permanently

The cement consumption notebook records bags used per structural element per date. On one entry, dated 24/07, the quantity field contains a handwritten question mark. The person recording did not know the number and it was never recovered. That data is gone forever.

Finding 4 — The tracker itself contains contradictions

We analyzed the master protocol tracker (84 rows). Two systematic problems:

Issue	Count	Implication
Rows showing progress above 100%, some reaching 154%	26 of 84 rows	The baseline denominator is wrong or unmaintained. The percentages are meaningless.

Rows with recorded non-conformances still marked as COMPLIED	17 rows affected	Either the sign-off criterion ignores the non-conformance column, or people are marking status without checking. Both are audit risks.
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Finding 5 — Procedures were copied from an unrelated project and never adapted

We reviewed three work procedure documents from the project's own procedures folder. All three have a correct cover page naming this project — and a body that still refers to a completely different project: a mining project executed by different companies in 2011–2012. Internal document codes still carry the other project's prefix.

► *This is a serious document-control finding in a state contract. It is also a product opportunity: generated documents that are always correctly attributed are a feature worth paying for.*

Finding 6 — They already have construction software, and quality still runs on WhatsApp

The project uses a construction management platform for site tasks, drawings and progress tracking. The quality manager shared the project link in the team group and asked everyone to install it. It is in active use for what it covers.

And yet every quality-critical exchange we observed in the team chat still flows through WhatsApp: test requests, slump photos, pour schedules, mix design reports, cement consumption figures, even formal memoranda. Two versions of the same schedule file were shared through the chat days apart.

This is not a story about a failed tool. It is a story about a gap. General construction management platforms handle tasks, plans and progress. They do not handle the quality protocol chain — the release sequence per layer, validation against the binding national standard, deferred cylinder strength results, non-conformance workflow, and the compiled quality dossier the state requires at closure. That work has nowhere to live, so it lives in WhatsApp and on paper.

Protokol does not replace their construction platform. It fills the quality compliance gap that no platform in this market covers.

Finding 7 — The quality system did not exist until three months ago

According to the Quality Specialist's own first monthly report, when he joined the project in June 2026 — with the project already nearly four years into execution — he found: no quality plan, no construction procedures implemented, no quality dossier, and no non-conformance register. Zero. He also identified physical defects in already-built structures: cracking in two retaining walls and segregation voids in several concrete walls.

5. THE OPPORTUNITY

5.1 Why a business exists here

Three forces converge:

- Legal obligation — quality documentation is mandatory, not discretionary. Under the Comptroller directive, registration carries sanctionable responsibility.
- Financial trigger — incomplete documentation delays payment certificates. There is immediate cash pain.
- Personal exposure — the quality engineer is individually liable for seven years. The buyer has skin in the game.

Most construction software sells efficiency. Protokol sells protection and cash flow. That is a fundamentally stronger sale.

5.2 Market size

Metric	Figure	Source
Stalled public works projects in Peru	2,741 as of early 2026	Comptroller General
Committed investment in stalled projects	Over S/ 67 billion	Comptroller General
Leading cause cited	Insufficient technical capacity in project management and documentation	Comptroller General
Audited projects that doubled cost and tripled schedule	32 in a single 2026 audit	Comptroller General
Our pilot project's own record	+156% budget, 10 time extensions	Project's internal monthly report

The relevant addressable market is every public infrastructure project executed under direct administration in Peru. Ayacucho region alone runs dozens simultaneously. The same regulatory framework applies in all 25 regions.

5.3 Competitive position — honest assessment

International construction management platforms exist and are available in Peru — our own pilot project uses one. Digital checklist products with mobile photo capture and signatures also exist in the Spanish-speaking market — Acofile Q (Spain) is one confirmed example, digitizing PPIs with offline capture, locked GPS evidence, and one-click non-conformance escalation. We are not the first to build digital site forms, and we should never claim we are. Acofile targets large EPC contractors in renewables, mining and infrastructure, mostly in Europe; it does not touch Peruvian public works or INFOBRAS, which is exactly our lane.

Three more players confirmed since: Calidad Cloud (Chile) is far larger, over 8,000 projects and 13,000 active users across Latin America, covering quality, productivity and warranty management with a Procore integration — but every named client is private construction or real estate, with no mention of public works anywhere on its site. ObraLink (Chile, also operating in Mexico and the UK) goes further technically, using computer vision and IoT sensors to measure concrete strength and progress without a human filling out a form — a plausible direction for the category long-term, not a Phase 0 concern. Bildin (Peru) is a local workforce-productivity tool, not a quality tool, and lists Calidad Cloud as a formal alliance partner — so Calidad Cloud already has a foothold in Peru indirectly, worth tracking even though neither touches administración directa or INFOBRAS today.

What none of them do is the compliance layer: encoding a specific project's binding technical criteria, enforcing the layer-by-layer release sequence, handling deferred lab results weeks after the pour, and compiling the quality dossier in the format the Peruvian state requires at contract closure. That is why our pilot project can run a construction platform and still manage all quality work through WhatsApp and paper.

Our defensibility is not the software. It is three things that compound:

Asset	Why it is hard to copy
Normative knowledge base	Each project's binding criteria are buried in its own technical file and lab mix designs. Extracting and structuring these correctly requires domain access. Every project we process makes the library deeper.
State system integration	Feeding INFOBRAS in the Comptroller's required format is a compliance feature, not a convenience feature. Foreign platforms will not build it.
Channel through credibility	Sector trust in Peru is personal. A civil engineer with institutional reputation opens doors that marketing spend cannot.

5.4 The wedge and the ladder

Stage	What we sell	To whom	Why they buy
Wedge	Quality record capture and dossier generation	Quality Specialist	Personal legal protection, stops chasing people
Step 2	Payment certificate backing package	Site Resident / contractor	Unblocks monthly cash flow
Step 3	Real-time compliance visibility and INFOBRAS feed	State entity / supervision	Audit readiness, sanction avoidance
Step 4	Multi-project quality intelligence across a region	Regional Government	Portfolio-level control of stalled-project risk

6. WHAT WE ARE BUILDING

Protokol captures quality data where the work happens, validates it against the legally binding standard, and produces the document package the state requires — automatically.

6.1 Three layers

Layer 1 — Capture (field)

A progressive web app. No install from an app store, no account creation, opens from a link, works on a low-end Android phone with intermittent signal. The field technician records what they just did in under five minutes.

- Guided forms per activity type, not free-form data entry
- Camera capture with GPS coordinates and tamper-resistant timestamp
- Offline-first: queues locally, syncs when signal returns
- Real-time validation feedback — the technician knows immediately if a value is out of range

Layer 2 — Intelligence (backend)

Maintains the true state of the project at all times and enforces the rules.

- Validates every submitted value against project-specific criteria loaded as data
- Automatically opens a non-conformance record when a value fails
- Tracks the release state of every panel across every protocol stage
- Manages the deferred workflow: cylinder strength results arriving 7 and 28 days after the pour
- Immutable evidence store — records are append-only with integrity hashing

Layer 3 — Output (office and state)

- Generated protocol PDF per record, in the format supervision accepts
- Live status view for the quality manager — what is released, pending, or non-conforming
- Quality dossier compilation for contract closure
- Export in INFOBRAS-compatible structure

6.2 What WhatsApp is and is not

WhatsApp is the notification and escalation channel — not the data capture channel. Notifications go out when a protocol completes, when a non-conformance opens, and when a record is overdue. Data entry happens in the PWA, because photo quality, GPS metadata and structured fields matter legally and WhatsApp degrades all three.

7. PHASE 0 — THE PILOT BUILD

► *Scope discipline: Phase 0 exists to answer one question, not to be a product.*

Does the field team record quality data through Protokol without the quality manager chasing them?

7.1 In scope

- PWA with guided forms for four activities: concrete, survey, compaction, steel (formwork is checked as part of the concrete pre-pour checklist, not as a separate activity)
- Backend API with the four endpoints in Section 8
- Immutable record storage with photo handling
- Project-specific validation criteria loaded as data
- Protocol PDF generation
- WhatsApp notifications to the quality manager
- Deferred cylinder result entry (7 and 28 day strength)
- Simple status view — a synced spreadsheet is acceptable for Phase 0

7.2 Explicitly out of scope

- Full web dashboard — deferred to Phase 1
- Native mobile app — the PWA is sufficient
- Multi-project support — one project, hardcoded, is fine
- INFOBRAS integration — design for it, do not build it yet
- Billing, subscriptions, self-service onboarding
- Complex authentication — see Section 9 on identity

8. API CONTRACTS

► *These contracts define what crosses the boundary. Everything inside is your call — stack, framework, patterns, deployment.*

8.1 POST /protocols

Submitted when the technician completes a form. Server validates against stored criteria and returns the verdict.

```
// Request
{
  "project_id": "AY-728-001",
  "device_token": "dvc_8f3a...", // issued once per device
  "technician_pin": "1234",
  "activity": "CONCRETE", // CONCRETE | SURVEY | COMPACTION | STEEL
  "recorded_at": "2026-09-11T08:47:00-05:00",
  "gps": { "lat": -13.1588, "lng": -74.2236 },
  "panel": "15",
  "chainage": "0+144",
  "measurements": {
    "slump_cm": 10,
    "mixer_id": "6D37",
    "delivery_note": "GR-00412", // legal delivery document
    "cylinders_cast": 4, // 4 per truck, confirmed 16 Sep 2026
    "design_fc": 210
  },
  "photo_ids": ["ph_a91c...", "ph_b22e..."],
  "notes": ""
}
```

```
// Response
{
  "protocol_id": "PRT-20260911-0847-015",
  "verdict": "PROVISIONAL_PASS", // PASS | PROVISIONAL_PASS | FAIL
  "checks": [
    { "field": "slump_cm", "expected": "8.9 - 12.7", "actual": 10, "result": "PASS" }
  ],
}
```

```

"nonconformance_id": null,
"pdf_url": "https://.../PRT-20260911-0847-015.pdf",
"pending": ["CYLINDER_7D", "CYLINDER_28D"]
}

```

► *PROVISIONAL_PASS matters: a concrete pour cannot be finally certified until cylinder strength results arrive weeks later. Modeling this as a two-stage verdict is a domain requirement, not a nicety.*

8.2 POST /protocols/{id}/cylinder-result

Submitted when the lab returns 7-day or 28-day compressive strength. Converts a provisional pass into a final verdict, or opens a non-conformance.

```

{
  "age_days": 28,
  "cylinder_code": "P-2026-0847-A",
  "strength_kgcm2": 231,
  "lab": "AKHISE",
  "report_photo_id": "ph_c77d..."
}

```

8.3 GET /projects/{id}/status

Returns the current release state of the whole project. Drives the status view and, later, the dashboard.

```

{
  "project_id": "AY-728-001",
  "as_of": "2026-09-11T09:15:00-05:00",
  "summary": {
    "total_expected": 84,
    "passed": 71,
    "provisional": 6,
    "failed": 3,
    "missing": 4
  },
  "protocols": [ /* ... */ ]
}

```

8.4 GET /projects/{id}/dossier

Compiles every protocol, photo, lab result and non-conformance into the quality dossier package.

```

{
  "dossier_url": "https://.../AY-728-dossier.pdf",
  "generated_at": "2026-09-11T09:30:00-05:00",
  "protocols_included": 84,
  "open_nonconformances": 1,
  "completeness_pct": 94.2
}

```

9. DATA MODEL AND IRREVERSIBLE DECISIONS

9.1 Core tables

Table	Purpose	Critical fields
projects	One row per construction project	id, name, contract_number, entity, execution_mode
protocols	The central artifact. One per recorded activity.	id, project_id, activity, recorded_at (with timezone), gps, panel, chainage, measurements (JSON), verdict, device_token, integrity_hash
criteria	Project-specific validation thresholds, loaded as data	project_id, activity, field, operator, min, max, source_reference
cylinders	Concrete test specimens and their deferred results	protocol_id, cylinder_code, cast_date, test_date, age_days, strength, lab
nonconformances	Failures requiring corrective	protocol_id, description, status, corrective_action, closed_at

	action	
photos	Evidence images with metadata	id, protocol_id, storage_key, gps, captured_at, hash
technicians	Field users	id, project_id, name, pin_hash, device_token, whatsapp, role
notifications	Outbound message log	recipient, event_type, payload, sent_at, status

9.2 Four decisions that cannot be undone later

Everything else in this build is reversible in days or weeks. These four are not. Please confirm your position on each before writing code.

Decision	Why it is irreversible	Our recommendation
Validation criteria in code vs. in database	If hardcoded, the entire product gets designed around single-project assumptions. Retrofitting touches everything.	Store as data from day one. Each project loads its own criteria. Non-negotiable for the business model.
Timestamp handling	Records stored without explicit timezone become legally ambiguous forever. Cannot be repaired retroactively.	Always store UTC with explicit offset. Display in project-local time.
Technician identity mechanism	The legal value of every record depends on proving who submitted it. Past records cannot be re-signed.	PIN for UX, plus a persistent device token issued once. Store device + PIN + GPS + integrity hash on every record.
Photo storage references	Thousands of photos with provider URLs embedded in immutable records make migration painful.	Store our own opaque identifier. Resolve to a URL at read time. Never persist a vendor URL inside a record.

9.3 Validation criteria — real values from the pilot project

These come from the project's binding technical file and the accredited lab's mix design. They load as data per project, never as constants.

Activity	Field	Pass criterion	Source
CONCRETE	slump_cm	8.9 - 12.7 cm (3.5" - 5"). Confirmed by the Quality Specialist, applies across elements, not per-partida	Quality Specialist, confirmed 16 Sep 2026
CONCRETE	design_fc	Per the activity's specified strength (140 / 175 / 210 / 245 / 280 kg/cm ²)	Technical file specifications
CONCRETE	cylinders_cast	4 cylinders per mixer truck (carmix), not per pour as a whole	Quality Specialist, confirmed 16 Sep 2026
CYLINDER	strength_kgcm2	≥ design f'c at 28 days	EG-2013
COMPACTION	compaction_pct	≥ 100% of Modified Proctor	Project quality plan
COMPACTION	moisture_deviation	Within ±1.5% of optimum	Project quality plan
COMPACTION	sub_base_thickness	≥ 20 cm	Technical file
COMPACTION	base_thickness	≥ 25 cm	Technical file
SURVEY	elevation_deviation	≤ 1 cm from design	Project quality plan
STEEL	bar_spacing_cm	Per structural drawing tolerance per bar diameter	Structural drawing
STEEL	concrete_cover_cm	Per structural drawing, minimum cover per element type	EG-2013 / drawing

► Update 16 Sep 2026: real thresholds above are confirmed by the Quality Specialist directly (slump range, cylinders per truck, f'c per partida). One operational nuance: a single pour can involve 20+ truck loads (guías de despacho), each with its own slump reading and its own set of 4 cylinders — design the concrete record to hold a list of truck entries, not a single slump/cylinder pair per protocol.

10. ACCEPTANCE CRITERIA

Written as testable statements. Each one is verifiable without interpretation.

#	Requirement
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R1	WHEN a technician submits a protocol with all required fields, THE SYSTEM SHALL persist an immutable record and return a verdict within 5 seconds under normal connectivity.
R2	WHEN a submitted measurement falls outside the configured criterion, THE SYSTEM SHALL create a linked non-conformance record and notify the configured recipient within 60 seconds.
R3	WHEN the device has no connectivity, THE SYSTEM SHALL queue the submission locally and sync automatically once connectivity returns, without user action.
R4	WHEN a concrete protocol is submitted, THE SYSTEM SHALL mark it PROVISIONAL_PASS and remain pending until cylinder results at 28 days are recorded.
R5	WHEN cylinder strength below design f _c is recorded, THE SYSTEM SHALL change the protocol verdict to FAIL and open a non-conformance.
R6	THE SYSTEM SHALL store every timestamp in UTC with explicit offset and never accept a client-supplied timestamp without recording server receipt time alongside it.
R7	THE SYSTEM SHALL reject any attempt to modify a persisted protocol record. Corrections occur through a new linked record, never by overwriting.
R8	WHEN a protocol is completed, THE SYSTEM SHALL generate a PDF containing all recorded values, photos, technician identity, timestamp and chainage.
R9	WHEN a photo is captured, THE SYSTEM SHALL store its GPS coordinates and capture time as metadata independent of the image file.
R10	WHEN an expected protocol has not been submitted 4 hours after its scheduled activity, THE SYSTEM SHALL notify the quality manager.

11. HOW WE WORK TOGETHER

Authority

- Stack, architecture, libraries, patterns, infrastructure: your call. Kenny does not have an opinion and will not develop one.
- What gets built, in what order, and what is in scope: Kenny's call. You can object on technical cost and that objection gets recorded.
- If we deadlock: the reversible option wins. If neither is reversible, we defer until we have more information.

Cadence

- Weekly deploy to a link Kenny can open on a phone. Whatever state it is in.
- Kenny tests within 48 hours and responds in writing: what works, what does not, what is missing.
- One 30-minute call per week. Fixed agenda: blockers, then decisions, then demo.

Domain questions

- Never guess about construction domain. If a decision depends on how the physical or legal process works, flag it as a DOMAIN BLOCKER and move to another task.
- Kenny responds to domain blockers within 24 hours, or within 48 if he needs to consult the field.
- Every domain answer gets written into this document, not just the chat. This document is the source of truth.

Decision records

Every architectural decision gets a short file in the repository: context, decision, consequences, and reversibility rating. Keeps both of us honest about what was decided and why.

Definition of done

Deployed and reachable by link · meets its acceptance criteria · Kenny tested it on a real phone · the architectural decision is recorded.

12. FUTURE DIRECTIONS SIGNALLED BY THE COMMERCIAL CHANNEL

► *None of this is Phase 0. It is recorded here so you are not designing in the dark about where this may go. The only thing I need from you on this section is a technical read, not extra scope.*

In the in-person meeting, David raised two directions he believes the sector is moving toward. He is our domain expert and commercial channel, so his read on where the market goes carries weight — but he is not setting technical direction, and neither am I. Both items below are yours to evaluate.

12.1 A retrieval-based assistant scoped to a single role

David proposed building a retrieval assistant aimed at one role only — the Quality Specialist, or alternatively the Site Resident — answering questions against that role's own normative and project documents. His argument is commercial, not technical: it would be cheaper and faster to deliver than full Protokol, could be sold sooner, and the revenue could fund the larger build.

Treat this as a possible separate product, not as a Protokol feature. It has a different scope, a different price point and arguably a different buyer. Whether we pursue it, and in what order relative to Phase 0, is a business decision I have not made yet. I am not asking you to scope or estimate it now.

12.2 Accumulating quality data across projects

The second direction is longer-range: as Protokol processes more projects, the accumulated body of normative criteria, protocol records and non-conformance history becomes an asset in its own right — potentially for cross-project analysis, benchmarking, or partnerships. Section 5.3 already names the normative knowledge base as one of our three defensibility assets; this is the same idea taken further.

Unlike 12.1, this one does touch your work today, but only as a constraint rather than a feature: whatever you build in Phase 0 should not make multi-project aggregation expensive to add later. Phase 0 stays single-project and hardcoded as stated in Section 7.2. The question is only whether the data model forecloses the option.

► *This connects directly to the first irreversible decision in Section 9.2 — validation criteria stored as data rather than code. That decision was already made for business-model reasons; it happens to also keep this door open.*

12.3 What I am asking of you here

Only one thing: tell me whether either direction should change how you design the Phase 0 data model. If the honest answer is that they should not, say so and we treat them as fully separate tracks. I would rather hear that than have you carry hypothetical requirements into a two-to-three week build.

13. WHAT I NEED FROM YOU

Read the document, then answer these. Nothing gets built until we have these settled.

#	Question	Why it matters
1	How many hours per week can you realistically give this alongside your other work?	Determines whether Phase 0 is 4 weeks or 10. I would rather plan honestly than miss.
2	Which parts of this document did not make sense?	The domain is foreign and I will have explained something badly. Tell me where.
3	Do you agree with the four irreversible decisions in Section 9.2, or would you do any of them differently? — ANSWERED 16 Sep 2026: agreed with all four as written.	These are the only ones we cannot fix later.
4	Anything in the acceptance criteria that is technically unreasonable or ambiguous?	Better to fix now than to build the wrong thing.
5	What infrastructure do you want, and what does it cost per month?	I need to budget it.

6	Is there something you would build first that I did not propose?	You may see a better sequencing than I do.
7	Do either of the future directions in Section 12 change how you would design the Phase 0 data model, or do we treat them as fully separate tracks? — ANSWERED 16 Sep 2026: RAG should not change Phase 0 architecture. The multi-project direction should only influence the relational model through project-scoped entities, no multi-project functionality now.	I need to know if today's decisions quietly foreclose something, not to add scope.